

WEEK 9:

COUNT DATA - POISSON REGRESSION

APPLIED STATISTICAL ANALYSIS II

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ROADMAP THROUGH STATS LAND

Where we've been:

- Over-arching goal: We're learning how to make inferences about a population from a sample
- Last time: We learned how to conduct a linear regression when our outcome is an (un)ordered category

Today we will:

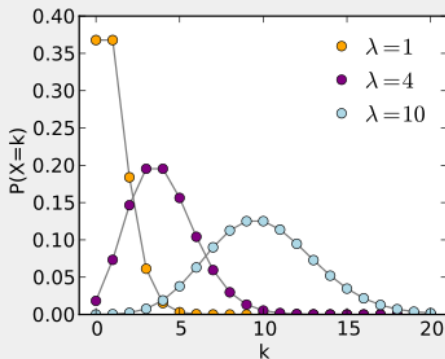
- Review exam
- Estimate & interpret a Poisson regression for count data! 😊

INTRODUCTION TO POISSON DISTRIBUTION

Let X be distributed as a Poisson random variable with single parameter λ

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad k \in (0, 1, 2, 3, 4, \dots)$$

- X is a discrete random variable with probabilities expressed in whole #s



INTRODUCTION TO POISSON DISTRIBUTION

- If $Y \sim \text{Poisson}(\lambda)$, then

$$E(Y) = \lambda \text{ and } \text{Var}(Y) = \lambda$$

- Mean and variance are equal, and variance is tied to mean
- If mean of Y increases with covariate X , so does variance of Y

FRAMEWORK: POISSON REGRESSION

Poisson regression model:

$$\ln(\lambda_i) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki}$$

where

$$\lambda_i = e^{\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki}}$$

- Poisson parameter λ_i depends on covariates of each observation
 - So, each observation can have its own mean
- Again, **mean depends on covariates, and variance depends on covariates**

BACKGROUND: POISSON REGRESSION

- Poisson regression is another **generalized linear model**
- Instead of a **log** function of Bernoulli parameter π_i (logistic regression), we use a **log** function of Poisson parameter λ_i

$$\lambda_i > 0 \rightarrow -\infty < \ln(\lambda_i) < \infty$$

BACKGROUND: POISSON REGRESSION

- The **logit** function in logistic model and **log** function in Poisson model are called the **link** functions for these GLMs
- In this modeling, we assume that $\ln(\lambda_i)$ is linearly related to independent variables
 - ▶ And that mean and variance are equal for a given λ_i
- An iterative process is used to solve the likelihood equations and get *maximum likelihood estimates* (MLE)
 - ▶ If you're interested in this specifically applied with Poisson, check out Gill (2001)

ZOOLOGY EXAMPLE: MATING OF ELEPHANTS

- There is competition for female mates between young and old male elephants¹
- Male elephants continue to grow throughout their lives → older elephants are larger and $\text{Pr}(\text{Successful mating}) \uparrow$



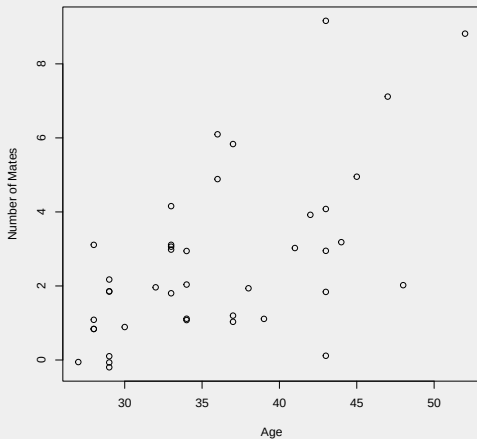
■ Variables:

- ▶ Response: # of mates
- ▶ Predictor: Age of male elephant (years)

¹Source: J. H. Poole, Mate Guarding, Reproductive Success and Female Choice in African Elephants, *Animal Behavior* 37 (1989): 842-49

ZOOLOGY EXAMPLE: MATING OF ELEPHANTS

Let's look at jitter scatterplot first



- It looks like the number of mates tends to be higher for older elephants
- Seems to be more variability in the number of mates as age increases
- Elephants of age 30 have between 0 and 4 mates
- Elephants of age 45 have between 0 and 9 mates

ZOOLOGY EXAMPLE: POISSON REGRESSION MODEL

- If dispersion (variance) $\uparrow$ with mean for a count response, then Poisson regression may be a good modeling choice
 - Why? Because variance is tied to mean!

$$\ln(\lambda_i) = \hat{\beta}_0 + \hat{\beta}_1 X$$

```
1 elephant_poisson <- glm(Matings ~ Age, data=elephant, family=poisson)
```

(Intercept)	-1.582** (0.545)
Age_in_Years	0.069*** (0.014)

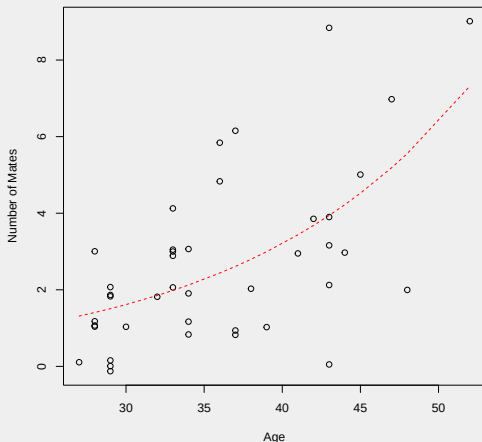
AIC	156.458
BIC	159.885
Log Likelihood	-76.229
Deviance	51.012
Num. obs.	41

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

EXAMPLE: POISSON REGRESSION CURVE

Add fitted curve to scatterplot:

```
1 coeffs <- coefficients(  
  elephant_poisson)  
2 xvalues <- sort(elephant$  
  Age)  
3 means <- exp(coeffs[1] +  
  coeffs[2]*xvalues)  
4 lines(xvalues, means, lty  
  =2, col="red")
```



Poisson regression is a nonlinear model for $E[Y]$

EXAMPLE: SIGNIFICANCE TEST

(Intercept)	-1.582** (0.545)
Age_in_Years	0.069*** (0.014)
AIC	156.458
BIC	159.885
Log Likelihood	-76.229
Deviance	51.012
Num. obs.	41

- Age is a reliable and positive predictor of # of mates for an elephant

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

EXAMPLE: PARAMETER INTERPRETATION

One covariate: $\ln(\lambda_i) = \beta_0 + \beta_1 X_i$

β_0 : e^{β_0} is mean of Poisson distribution when $X = 0$

β_1 : Increasing X by 1 unit has a multiplicative effect on the mean of Poisson by e^{β_1}

$$\frac{\lambda_{(x+1)}}{\lambda_{(x)}} = \frac{e^{\beta_0 + \beta_1(x+1)}}{e^{\beta_0 + \beta_1 x}} = \frac{e^{\beta_0} e^{\beta_1 x} e^{\beta_1}}{e^{\beta_0} e^{\beta_1 x}} = e^{\beta_1}$$

$$\lambda_{(x+1)} = \lambda_{(x)} e^{\beta_1}$$

- If $\beta_1 > 0$, then expected count increases as X increases
- If $\beta_1 < 0$, then expected count decreases as X increases

EXAMPLE: PARAMETER INTERPRETATION

For the elephant data:

$\hat{\beta}_0$: No inherent meaning in the context of the data since $age = 0$ is not meaningful, outside of range of possible data

- Since coefficient is positive, expected # of mates $\uparrow$ with age

$\hat{\beta}_1$: An increase of 1 year in age increases expected number of elephant mates by a multiplicative factor of $e^{0.06859} \approx 1.07$

EXAMPLE: GETTING FITTED VALUES

- Fitted model:

$$\lambda_i = e^{\hat{\beta}_0 + \hat{\beta}_1 X_i}$$

- What is fitted count for an elephant of 30 years?

```
> lambda30<-exp(coeff[1]+coeff[2]*30)
> lambda30
(Intercept)
1.614098
```

- Estimated mean number of mates = 1.6
- Estimated variance in number of mates = 1.6

EXAMPLE: ESTIMATING FITTED VALUES

$$\lambda_i = e^{\hat{\beta}_0 + \hat{\beta}_1 X_i}$$

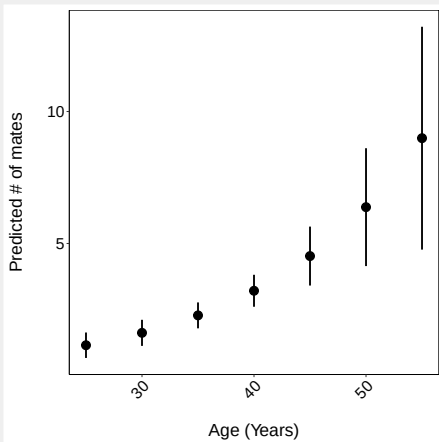
- What is fitted count for an elephant of 45 years?

```
> lambda45<-exp(coeff[1]+coeff[2]*45)
> lambda45
(Intercept)
  4.522968
```

- Estimated mean number of mates = 4.5
- Estimated variance in number of mates = 4.5

GETTING FITTED VALUES IN R

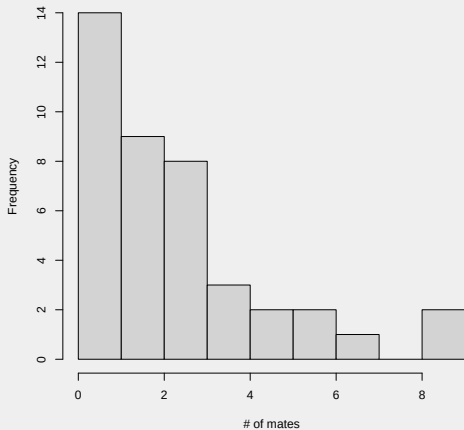
```
1 predicted_values <- cbind(predict(elephant_poisson, data.frame(Age = seq(25, 55, 5)),  
                             type="response", se.fit=TRUE), data.frame(Age = seq(25, 55, 5)))  
2 # create lower and upper bounds for CIs  
3 predicted_values$lowerBound <- predicted_values$fit - 1.96*predicted_values$se.fit  
4 predicted_values$upperBound <- predicted_values$fit + 1.96*predicted_values$se.fit
```



ASSUMPTIONS: OVER-DISPERSION

- Assuming that model is correctly specified, assumption that conditional variance is equal to conditional mean should be checked
- There are several tests including the likelihood ratio test of over-dispersion parameter ϕ by running same model using negative binomial distribution
- R package AER provides many functions for count data including `dispersiontest` for testing over-dispersion
- One common cause of over-dispersion is excess zeros, which in turn are generated by an additional data generating process
- In this situation, zero-inflated model should be considered

ZERO INFLATED POISSON: # OF MATES



- Though predictors do seem to impact distribution of elephant mates, Poisson regression may not be a good fit (large # of os)
- We'll check by
 - ▶ Running an over-dispersion test
 - ▶ Fit a **zero-inflated Poisson regression**

OVER-DISPERSION TEST IN R

```
1 # check equal variance assumption
2 dispersiontest(elephant_poisson)
```

Overdispersion test

```
data:  elephant_poisson
z = 0.49631, p-value = 0.3098
alternative hypothesis: true dispersion is greater than 1
sample estimates:
dispersion
1.107951
```

Doesn't seem like we really need a ZIP model, but we'll do it anyway...

INTUITION BEHIND ZERO-INFLATED POISSON

- In terms of fitting the model, we combine **logistic regression model** and **Poisson regression model**
- ZIP model:
 - ▶ We model probability of being a perfect zero as a logistic regression
 - ▶ Then, we model Poisson part as a Poisson regression
- There are two generalized linear models working together to explain data

ZIP MODEL IN R

R contributed package “pscl” contains the function *zeroinfl*:

```
1 # same equation for logit and poisson
2 zeroinfl_poisson <- zeroinfl(Matings ~ Age, data=elephant, dist="poisson")
```

Count model: (Intercept)	-1.45** (0.55)
Count model: Age_in_Years	0.07*** (0.01)
Zero model: (Intercept)	222.47 (232.27)
Zero model: Age_in_Years	-8.12 (8.44)
AIC	157.88
Log Likelihood	-74.94
Num. obs.	41

Further evidence we don't really need zero-inflated model

EXPOSURE VARIABLES: OFFSET PARAMETER

- Count data often have an exposure variable, which indicates # of times event could have happened
- This variable should be incorporated into a Poisson model using `offset` option

EX: FOOD INSECURITY IN TANZANIA AND MOZAMBIQUE

- Survey data from households about agriculture
- Covered such things as:
 - ▶ Household features (e.g. construction materials used, number of household members)
 - ▶ Agricultural practices (e.g. water usage)
 - ▶ Assets (e.g. number and types of livestock)
 - ▶ Details about the household members
- Collected through interviews conducted between Nov. 2016 - June 2017 using forms downloaded to Android Smartphones

WHAT PREDICTS OWNING MORE LIVESTOCK?

■ **Outcome:** Livestock count [1-5]

■ **Predictors:**

- ▶ # of years lived in village
- ▶ # of people who live in household
- ▶ Whether they're apart of a farmer cooperative
- ▶ Conflict with other farmers

OWNING LIVESTOCK: ESTIMATE POISSON REGRESSION

```
1 # load data
2 safi <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/StatsII_2026/main/datasets/SAFI.csv",
  stringsAsFactors = T)

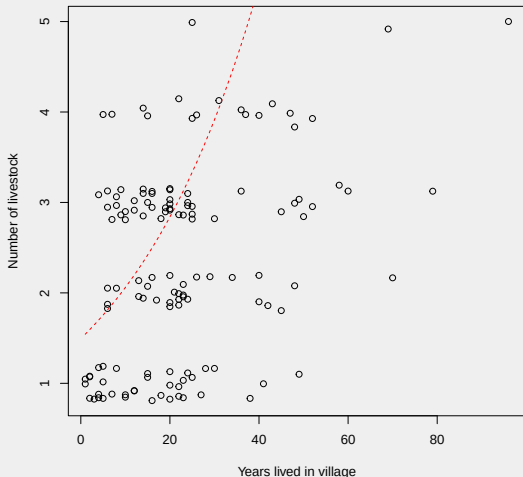
1
2 # estimate poisson regression model
3 safi_poisson <- glm(liv_count ~
  no_membrs + years_liv +
  memb_assoc + affect_
  conflicts, data=safi,
  family=poisson)
```

(Intercept)	0.40**
	(0.15)
no_membrs	0.03
	(0.02)
years_liv	0.01*
	(0.00)
memb_assoc_yes	-0.03
	(0.16)
affect_conflicts_frequently	0.09
	(0.24)
affect_conflicts_more_once	0.14
	(0.15)
affect_conflicts_once	0.09
	(0.25)
AIC	417.98
BIC	438.11
Log Likelihood	-201.99
Deviance	54.52
N	131

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

OWNING LIVESTOCK: POISSON REGRESSION CURVE

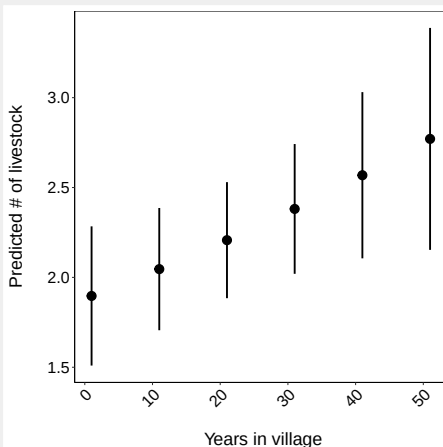
Add fitted curve to scatterplot:



As # of years in village $\uparrow$, $\uparrow$ expected # of livestock

OWNING LIVESTOCK: FITTED VALUES IN R

```
1 safi_ex <- data.frame(no_membrs = rep(mean(safi$no_membrs), 6),  
2                           years_liv = seq(1, 60, 10),  
3                           memb_assoc = rep("no", 6),  
4                           affect_conflicts = rep("never", 6))  
5 pred_safi <- cbind(predict(safi_poisson, safi_ex, type="response", se.fit=TRUE), safi_ex)
```



OWNING LIVESTOCK: OVER-DISPERSION

```
1 dispersiontest(safi_poisson)
```

Overdispersion test

```
data: safi_poisson  
z = -12.433, p-value = 1  
alternative hypothesis: true dispersion is greater than 1  
sample estimates:  
dispersion  
0.4130252
```

Don't really need a ZIP model

EX: AIR POLLUTANTS AND PRE-MATURE MORTALITY

National Mortality, Morbidity, and Air Pollution Study (NMMAPS), includes info on daily mortality, air pollution, and weather

- Data from 108 U.S. cities for 14 years (1987–2000)
- Covered such things as:
 - ▶ Health outcomes
 - ▶ Meteorological conditions
 - ▶ Environmental pollutants (PM_{10} , $PM_{2.5}$, SO_2 , O_3 , NO_2 , and CO)

WHAT PREDICTS RESPIRATORY DEATHS?

■ **Outcome:** Respiratory deaths

■ **Predictors:**

- ▶ temp: Mean temperature (in Celsius)
- ▶ rhum: Mean relative humidity
- ▶ pm10: Median particles in 2.5-10 per cubic meter
- ▶ o3: Ozone in parts per billion

RESPIRATORY DEATHS: ESTIMATE POISSON REGRESSION

```
1 NMMAPS <- read.csv("https://raw
  .githubusercontent.com/ASDS
  -TCD/StatsII_2026/main/
  datasets/NMMAPS.csv",
  stringsAsFactors = T)

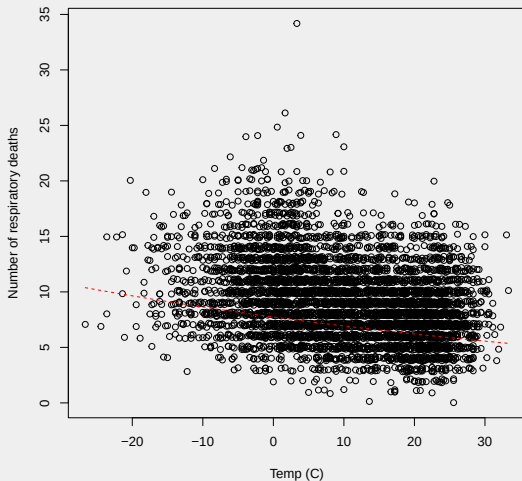
1 NMMAPS_poisson <- glm(resp ~
  temp + rhum + pm10 + o3 +
  as.factor(year), data=
  NMMAPS, family=poisson)
```

(Intercept)	2.0464*** (0.0396)
temp	-0.0110*** (0.0006)
rhum	0.0018*** (0.0004)
pm10	0.0011*** (0.0003)
o3	0.0015* (0.0007)
Year fixed-effect	✓
AIC	19757.7585
BIC	19851.3127
Log Likelihood	-9863.8792
Deviance	4628.9970
N	3778

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

RESPIRATORY DEATHS: POISSON REGRESSION CURVE

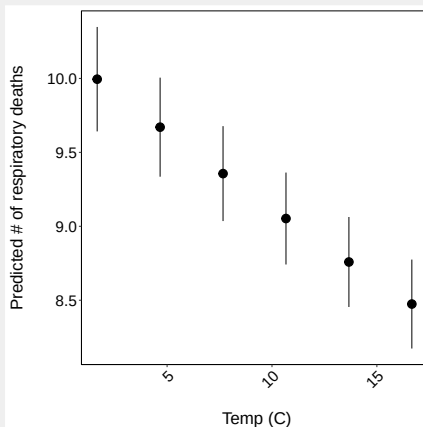
Add fitted curve to scatterplot:



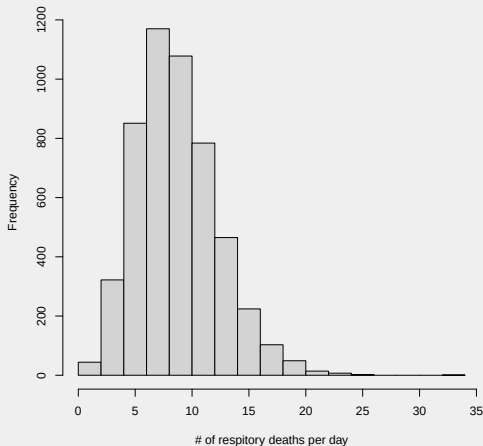
As average daily temp $\uparrow$, $\downarrow$ expected # of respiratory deaths

RESPIRATORY DEATHS: FITTED VALUES IN R

```
1 NMMAPS_ex <- data.frame(temp = seq(1.667, 19.444, 3),  
2                           rhum = rep(mean(NMMAPS$rhum, na.rm = T), 6),  
3                           pm10 = rep(mean(NMMAPS$pm10, na.rm = T), 6),  
4                           o3 = rep(mean(NMMAPS$o3, na.rm = T), 6),  
5                           year = rep("1994", 6))  
6 pred_NMMAPS <- cbind(predict(NMMAPS_poisson, NMMAPS_ex, type="response", se.fit=TRUE),  
  NMMAPS_ex)
```



ZERO INFLATED POISSON: # OF DEATHS



- Doesn't appear to be over-dispersion
- But, we'll still check by
 - ▶ Running an over-dispersion test
 - ▶ Fit a **zero-inflated Poisson regression**

RESPIRATORY DEATHS: OVER-DISPERSION

```
1 dispersiontest(NMMAPS_poisson)
```

Overdispersion test

data: NMMAPS_poisson

z = 6.9628, p-value = 1.668e-12

alternative hypothesis: true dispersion is greater than 1
sample estimates:

dispersion

1.219066

Probably don't need a ZIP model

ZIP MODEL IN R

```
1 zeroinfl_NMMAPS <- zeroinfl(resp ~ temp + rhum + pm10 + o3, data=NMMAPS, dist="poisson")
```

Count model: (Intercept)	2.1281*** (0.0350)
Count model: temp	-0.0111*** (0.0006)
Count model: rhum	0.0018*** (0.0004)
Count model: pm10	0.0011*** (0.0003)
Count model: o3	0.0017* (0.0007)
Zero model: (Intercept)	-12.6779 (9.2424)
Zero model: temp	0.1912 (0.3143)
Zero model: rhum	-0.1189 (0.1397)
Zero model: pm10	-0.1430 (0.1191)
Zero model: o3	0.3233 (0.2014)
AIC	19807.4942
Log Likelihood	-9893.7471
Num. obs.	3778

Further evidence we don't really need zero-inflated model

CASE STUDY: HOUSEHOLD SIZE IN THE PHILIPPINES

- Outcome: $Y = \#$ of household members *other than* the head
- Data from 2015 Family Income and Expenditure Survey (FIES), 1500-household subset from 5 regions
 - ▶ Regions: Central Luzon, Metro Manila, Ilocos, Davao, Visayas
- Key question: At what age are heads of household most likely to have the largest household size?
- Secondary question: Is the age-household size relationship similar for poorer households (proxied by roof material)?

VARIABLES AND DATA ORGANIZATION

- Each row: One household at survey time
- Covariates:
 - ▶ location: Region (5 categories)
 - ▶ age: Age of head of household
 - ▶ roof: Roof type
 - Predominantly light/salvaged material
 - Predominantly strong material (proxy for greater wealth)
- Outcome is a count, so Poisson regression is a natural starting point

FIT POISSON MODEL

```
1 full_model <- glm(total ~ age + location + roof, family =  
  poisson, data = fHH1)  
2  
3 summary(full_model)
```

$$\log(\hat{\lambda}_i) = 1.552 - 0.0047 \text{ age}_i + 0.061 \text{ Ilocos}_i + 0.054 \text{ Metro Manila}_i \\ + 0.112 \text{ Visayas}_i - 0.019 \text{ Davao}_i + 0.012 \text{ light/salvaged roof}_i$$

MODEL COEFFICIENTS AND RATE RATIOS

- **Age:** $\hat{\beta}_1 = -0.0047$, so $\exp(-0.0047) \approx 0.995$
 - ▶ Each extra year: 0.5% *decrease* in expected household size
 - ▶ 95% CI for rate ratio: (0.993, 0.997)
- **Location example:** Visayas vs Central Luzon = 0.112
 - ▶ $\exp(0.112) \approx 1.12$: 12% *larger* households in Visayas
- **Roof:** Light/salvaged vs strong roofs = 0.012
 - ▶ Little evidence of wealth effect after adjusting for age/location ($\approx 1\%$)

PREDICTED VALUES

```
1 new_data <- expand.grid(age = seq(25, 65, by = 5),  
2   location = levels(fHH1$location),  
3   roof = levels(fHH1$roof))  
4  
5 pred <- predict(full_model, newdata = new_data,  
6 type = "response")
```

- Age 40, Central Luzon, strong roof: $\hat{\lambda} \approx 3.8$
- Age 40, Visayas, strong roof: $\hat{\lambda} \approx 4.2$ (10% higher)
- Age 60, Central Luzon, strong roof: $\hat{\lambda} \approx 3.2$
- Age 60, Visayas, strong roof: $\hat{\lambda} \approx 3.5$

OVER-DISPERSION

```
1 dispersiontest(full_model)
2
```

```
Overdispersion test
data: full_model
z = 8.234, p-value < 2.2e-16
alternative hypothesis: true lambda > 1
```

- Evidence of moderate over-dispersion

ANOTHER POSSIBLE SOLUTION: QUASI-POISSON

- We don't have an issue with zero inflation in this case
- Fits same mean structure as Poisson but allows dispersion parameter $\phi \neq 1$
- Standard errors are scaled by $\sqrt{\phi}$ (wider CIs if overdispersed)
- Coefficients and predictions unchanged; inference adjusted

```
1 mod_quasi <- glm(total ~ age + location + roof,  
2     family = quasipoisson(link = "log"),  
3     data = fHH1)  
4  
5 summary(mod_quasi) # note wider SEs compared to Poisson  
6 coef(summary(mod_quasi)) # p-values based on adjusted SEs  
7
```

ESTIMATED COEFFICIENTS: QUASI-POISSON

	$\hat{\beta}$	SE	p-value
(Intercept)	1.552	(0.038)	4.72***
age	-0.0047	(0.0011)	0.995***
Ilocos Region	0.061	(0.052)	1.06*
Metro Manila	0.054	(0.049)	1.06*
Visayas	0.112	(0.050)	1.12**
Davao Region	-0.019	(0.053)	0.98
Light/Salvaged Roof	0.012	(0.041)	1.01

PRACTICAL RECOMMENDATIONS

- **Predictions:** Still valid from original Poisson model (mean structure unchanged)
- **Moderate overdispersion ($\phi \approx 1.5$):** Use quasipoisson
- **Severe overdispersion ($\phi > 3$) and/or many os:** Consider negative binomial

WRAP UP

In this lesson, we went over how to...

- Estimate and interpret a Poisson regression for count data

Next time, we'll talk about...

- Duration models
- Censoring & truncation
- Selection